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RESEARCH ARTICLE

Early MTS Forecasting for Dynamic Stock Prediction: A Double Q-Learning Ensemble Approach

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ABSTRACT Multivariate time series (MTS) forecasting is a rapidly expanding field with diverse and futuristic applications. However, traditional statistical learning models need more prediction accuracy when faced with dynamic variability, non-linearity, and non-stationarity, as well as the challenge of selecting MTS data for classification. Moreover, the existing methods for early classification of multivariate time series data suffer from numerous severe challenges, including evaluating the length of testing the MTS data component, which must be equal to the training MTS data component, and the availability of faulty data components in MTS. To address this issue, we propose a novel framework for early MTS forecasting using double Q-learning-based ensemble techniques to improve prediction accuracy. Our framework uses Q-learning agents to select optimal actions, which results in maximum rewards and accurate prediction. We investigate the ensemble behavior of learned agents using double Q-learning and Gaussian Process Classifiers (GPC) for early forecasting of MTS data. We also determine the minimum required time-series length for classifying faulty data components using the probabilistic Auto-Regressive Integrated Moving Average (ARIMA) model, enhancing framework robustness and mitigating miss-classification accuracy. Our proposed framework achieves 99.89% accuracy for early forecasting, surpassing existing methods based on different benchmark settings and publicly available multivariate time-series datasets. The framework provides a promising solution to the challenges of accurate MTS forecasting and offers insights into the early prediction of recent stock market trading data.

INDEX TERMS Q-learning, ensemble learning, Markov decision processes, classification, dynamic stock marketing, PPO, A2C, DDPG.

I. INTRODUCTION

Time series classification (TSC) is a crucial area of research in the Machine Learning (ML) community due to the prevalence of time series across various futuristic applications [1]. A time series data consists of sequentially a collection of observations about any item in a given time interval, which corresponds to an ordered succession of sensory data points over a set period. Assume that the incidence of entity activity is observed using several factors through sensor capability that provide a series of multivariate data points [3], called a Multivariate Time Series (MTS) [3]. The models are built based on the training MTS datasets and employed to classify new MTS data components in MTS classification. The futuristic applications of MTS include healthcare monitoring and predictive analytics of patients based on acquired multiple physiological sensor data. The health sensors produce temporal sequences of

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multiple observations of human activity (eating, sleeping, running, resting, and many more). TSC can be divided broadly into two categories: (1) univariate time series data classification and (2) multivariate TSC, depending on whether evaluations from single variables or multiple depending variables are considered in time series classification tasks [3], [4].

Univariate time series classification refers to categorizing and assigning discrete class labels to individual data sequences that contain a single variable observed over time. Meanwhile, MTS classification techniques only predict the class label of unknown MTS data when fully observed over a given time. Conventional MTS classification techniques accurately predict an unknown MTS's class labels only when the MTS is properly annotated with a class label and is fully observed. The primary aims of MTS classification systems are to achieve a discriminatory Minimum Required Length (MRL) of multivariate data using training datasets while maintaining a high level of model accuracy. MRL consists of discriminatory sets or a minimum number of MTS data points needed to accurately predict optimal data points for MTS forecasting before data processing, along with the complete model execution. Statistical machine learning techniques are widely used in multivariate time series classification to solve the problem of optimal data prediction. However, due to the availability of faulty data components and the minimum length required for training and testing data in MTS, the models cannot provide high-level accuracy for accurate prediction.

Some shallow approaches have tried to address this issue through temporal time series analysis [4]. However, these methods may need to accurately represent the multivariate time series, leading to overestimated, oversimplified data and less accurate results. Although supervised ML-based methods, such as evolutionary-based methods, do not require the computation of any parameter for the number of discretization intervals for early classification, models still perform the earlier classification of MTS based on the discretization process separately before model training, which requires additional time to perform [4]. MTS categorization has received substantial attention in recent decades for early categorization of time series employing Deep Learning (DL) approaches, and it has demonstrated promising results in this sector. Karim et al. [3] introduced two DL frameworks using MALSTM-FCN techniques for early classification. Furthermore, the MLSTM-FCN model was used to improve the ALSTM-Fully Convolution Network (FCN) approaches for early time series data categorization using LSTM-Fully Convolution Network (FCN). By adding distinct squeeze and excitation blocks, the suggested framework improved the classification performance of early classification of multivariate time series data [5]. Recently, Khan et al. [6] introduced a model that uses Recurrent Neural Network (RNN) variants instead of the Long Short-Term Memory (LSTM) units in the deployed MALSTM-FCN model and integrates MLSTM-FCN along with the SE and Fully Convolution Network (FCN) blocks to increase the overall performance of the proposed system.

To solve this problem, interdisciplinary researchers have experimented with Reinforcement Learning (RL) and ML techniques to analyze multivariate stock market prices in real-time. The authors of [4] used to analyze MTS data for accurate prediction of trading data, resulting in improved accuracy, efficiency, and speed of decision-making processes, leading to a significant increase in demand for reinforcement learning techniques, especially in the financial industry [4]. The reinforcement learning paradigm has revolutionized how we approach solving complex problems, and its potential applications are endless. As more businesses and industries continue to explore this approach, we can expect to see even more exciting breakthroughs in the future [4], [5], [6], [7]. Unlike traditional supervised learning, RL offers several advantages, including handling complex, high-dimensional input spaces and adapting to dynamic environments [2]. Conventional methods often require extensive feature engineering, which can be time-consuming and may not capture the full complexity of the data [2]. Also, RL shows promise in addressing these challenges, but it has limitations, such as being computationally expensive, especially with large state or action spaces [2]. Furthermore, training RL agents for classification tasks requires careful design choices, such as reward shaping and exploration strategies, to ensure convergence and generalization. Another challenge faced by RL-based classification methods is scalability. While RL algorithms have succeeded in small-scale environments, scaling them to handle large and high-dimensional datasets remains a significant challenge. To solve these problems, we proposed a novel framework to accurately predict multivariate time series using the double Q-learning technique.

A. MOTIVATION AND CONTRIBUTIONS

The profitable maneuvers in a complex and dynamic stock market make it more challenging to design accurate MTS classification systems and to predict multivariate trade data points [7], [8]. Therefore, the conventional method is first applied to measure the expected stock return. The information of the local stock data is accomplished into clustering patterns using ML and fuzzy modelling techniques [7], [8], [9]. Existing methods for Multi-Time Series (MTS) forecasting in dynamic stock prediction primarily rely on single-agent reinforcement learning algorithms, such as Q-learning , for multivariate time series classification. Therefore, a robust MTS classification system must be designed for accurate market prediction [11], [12], [13], [14]. The proposed framework identifies missing or faulty components of MTS data to improve the overall accuracy of the proposed framework [8]. Hence, the proposed framework better representations of faulty and non-faulty data components of multivariate time series data using partially ordered set (POSET)–based Hasse representations using mathematical modelling (shown in Fig 2). The data representations provide for finding covariance relationships between faulty and non-faulty data components for evaluations [10], [11], [12], [13], [14], [15]. The nodes of the Hasse representational-based graphs are presented in color codes, with blue nodes depicting the most relevant data components and red nodes showing faulty or irrelevant data components. The Gaussian process classifier (GPC) is then used to classify new data components of The multivariate time series as faulty or relevant for early predictions.

To evaluate the discriminatory components from the data set of all components, this study utilizes unsupervised learning (k-means clustering technique) to measure the relevancy score of the components. The relevancy score of a component $[C_i]$ is measured as $\mathcal{S}(\mathbf{C}_i) = \mathcal{A}_\mathbf{D} - \mathcal{A}_{\mathbf{D}_{-i}}$, where $\mathcal{A}_\mathbf{D}$ and $\mathcal{A}_{\mathbf{D}_{-i}}$ shows the accuracy values employed by applying k-means clustering on subset datasets $[D]$ and $[D_i]$, respectively. The proposed framework employs k-means clustering correlation-based similarity measure (euclidean distance-based measure similarity) to measure the nearest cluster during clustering. Moreover, the represents provide for finding covariance relationships between faulty and non-faulty data components for better evaluation of classification of multivariate time series predictions. The nodes of the Hasse graphs are presented in color codes, with white nodes depicting the most relevant data components and red nodes showing faulty or irrelevant data components. The GPC is then used to classify new data components as faulty or relevant for early predictions.

The major contributions in this work are illustrated as follows: The major contributions in this work are illustrated as follows:

1) We propose a novel framework for early forecasting of MTS data. This framework uses ensemble techniques based on the double Q-learning method to deal with the problems of dynamic variability, non-linearity, and non-stationarity , as well as choosing MTS data for classification. By proposing this framework, the research provides a new approach to MTS forecasting that improves prediction accuracy and provides valuable information for decision-makers in the finance sector and beyond.
2) The framework's use of double Q-learning agents to select optimal actions leads to maximum rewards and enhanced prediction accuracy. By leveraging ensemble techniques and GPC, this study investigates the ensemble behaviour of learned agents of Q-learning, resulting in improved forecasting performance. The contribution lies in achieving remarkable accuracy for early forecasting, surpassing existing methods and benchmark settings across various publicly available time-series datasets.
3) Next, we determine the minimum required length (MRL) of multivariate time series for classifying class labels of faulty data components using the probabilistic Auto-Regressive Integrated Moving Average (ARIMA) model. This analysis enhances framework robustness and reduces miss-classification accuracy, contributing to more reliable MTS forecasting.
4) The proposed framework offers insights into the early prediction of recent stock market trading data. By accurately forecasting stock market trends early, the research provides valuable information for financial decision-makers, enabling proactive decision-making and risk mitigation.

B. PAPER STRUCTURE

The remainder of this is organized as follows. The proposed framework is given in Section III. Section IV depicts the proposed system's experimental results, the performance evaluation, and comparative performance analysis with existing methods. Finally, the conclusion and future research directions are given in Section VI.

II. PRELIMINARY

This section describes the definitions, notations, and symbolic representations.

1) **Deep Q-Learning Network (DQN):** DQN-based algorithm efficiently classifies stock predictions using multivariate time series data. It performs an approximation on optimal Q-function DQ^* with parameters $[O_t, a_t, r_t, O_{t+1}]$ between an agent and its environment. DQN estimates the loss of the models based on the gradient descent step, measured loss function (shown in Eq. 1 and Eq. 2).

$$L_t(\mathbf{\Theta}_t) = \left(r + \gamma \max_{a'} DQ\left(\mathbf{O}', a', \mathbf{\Theta}_t^-\right)\right) - A \quad (1)$$

$$A = \left(DQ\left(\mathbf{O}, a, \mathbf{\Theta}_t\right)\right)^2 \quad (2)$$

Using Θ_t^- to denote an older, unchanging version of the Deep Q-Network (DQN), the agent gradually learns to estimate the connections between being in a state $\mathbf{O}_t$, selecting an action a_t, and receiving future rewards g_t.

2) **Univariate time series classification:** Traditional ML models are highly used based on statistical measures and class labels of the univariate time series data [X] = [X1, X2, $\cdots$, X_T]. Data [x] is an ordered set of real stock values with length [T].
3) **Multivariate time series(MTS) classification:** MTS classification is depicted as For a given training dataset [X] with labelled class labels $[L_1 L_2, L_l$, the task of MTS classification is to learn a mapping function H: $R^{n\times M}$ where class labels (l) $\in$ L. The examples of early MTS classification tasks are the (1) classification of human activity based on collected health data through different physiological sensor data, (2) different smart appliances may be going through different kinds of faults during their life cycle. These faults may be recognized by predicting the multivariate time series (MTS) generated by the connected sensors belonging to this fault.
4) **MTS dataset:** A set of pairs (X_i, Y_i) forms a dataset D, where $[X_i]$ may be a univariate or multivariate time series and Y_i is its corresponding one label vector. The one-hot

label vector Y_i is a K-length vector, where each element $j \in [1, K]$ equals 1 if the class of X_i is j and 0 otherwise. The dataset D contains [N] pairs and can contain data from K's different classes.

5) The observations provided to the agent at time step t, denoted as $\mathbf{O}_t$, consist of a partial time series collected up to time t:

$$\mathbf{O}t = \mathbf{S}\colon t \tag{3}$$

The agent's available actions denoted as a_t, include either waiting for additional observations or predicting a class label from the set $\mathcal{L}$:

$$a_t \in \mathcal{A},\ \text{where } \mathcal{A} = \text{wait}, \bigcup_{k\in\mathcal{L}} \text{predict label } k \tag{4}$$

The agent selects an action a_t based on its policy π_Θ parameterized by Θ, such that $a_t = \pi_\Theta\,(\mathbf{O}_t)$:

$$\pi_\Theta : \mathbf{S}_{it}, \forall t \in [1, T] \rightarrow \mathcal{A} \tag{5}$$

If the action at time step t, denoted as a_t, corresponds to waiting, the episode continues, and the subsequent observation is the partial time series extended by one additional point: $\mathbf{O}t + 1 = \mathbf{S}t + 1 + 1$. However, if the action a_t belongs to the set of predicting labels $\bigcup_{k\in\mathcal{L}}\cdot$ predict label k, the episode terminates. An episode concludes either when the partial sequence is fully observed, $\mathbf{O}t = \mathbf{S}$, or when the agent predicts a class label, $a_t \in \bigcup k \in \mathcal{C}$ predict label k. Rewards r_t play a crucial role in the agent's training process, as they encode the agent's task and significantly influence its behaviour. Rewards must align with the early classifier objective: providing swift predictions while maintaining acceptable accuracy.

6) Based on the deterministic policy, the double Q-learning model is used based on improve ed Q-value as shown in Eq. 12-13). Therefore, once we computed increased Q-values, the policy was improved to calculate the maximum rewards (R) (shown in Eq. 12).

$$R = argmax_a E\left[\sum_{t=0}^{\infty} \gamma^i \times R_{\pi(st)}(s_t, s_{t+1})\right] \tag{6}$$

$$Q_\pi(s_t, a_t) = \epsilon_{s_{t+1}}[r(s_t, a_t, s_{t+1}) + CC \tag{7}$$

$$CC = [\gamma\epsilon_{a_{t+1}\sim\pi(s_{t+1})}[Q_\pi(s_{t+1}, a_{t+1})]] \tag{8}$$

where the convergence $[0 \le \gamma \le 1]$ and policy evaluation computed the state value function $(V^\pi(s_t))$.

7) An optimal policy $(\pi : S \rightarrow A)$ (here S denotes the set of states$[(s_1), \cdots (s_t)]$ and A is the set of actions $[(a_1), \cdots (a_t)]$) is a policy is maximized the possible reward from a state (s), called value, $(V^\pi(s_t))$ for all states. Equation 9 shows a typical definition of this:

$$V^\pi(s_t) = r_{t+1} + \gamma r_{t+2} + \gamma^2 r_{t+3} + \ldots = \sum_{k=0}^{\infty} \gamma^k r_{t+k+1} \tag{9}$$

Here, γ $(0\le\ \gamma\ \le 1)$ is a tuning constant factor that measures the relative value of delayed versus immediate rewards.

8) $V^\pi(s)$ shows the possible cumulative reward obtained by following an arbitrary policy (π) from an arbitrary initial state $(s_0$. Then an optimal policy, (π) is derived as follows:

$$\pi^* = \arg\max_\pi V^\pi(s), (\forall s) \tag{10}$$

III. PROPOSED SYSTEM

The proposed framework consists of several steps: (1) input multivariate time series (MTS), which is a collection of an ordered set of data points. Let us consider that [C] depicts an MTS composed of n time series. Such n time series are called as [n] MTS data components of the multivariate time series. The i^{th} component of [C] is shown by symbolic notation $[C_i]$, where $[1 \le i \le n]$, (2) data description and pre-processing of MTS, (3) model building using GYM environment is an open source python library and (4) model selection and training step for forecasting algorithms, (5) combining models using ensemble strategy for building fusion model, (6) classification evaluation of relevancy scores for MTS for accurate classification((shown in Fig. 1).

The proposed framework used different agents (1,2,$\cdots$, N) to learn the stochastic behaviour of a volatile market environment using the Markov Decision Process (MDP) utilizing the state and actions of agents. The agent performs the optimal set of actions to learn chaotic natures based on measured expected values of state value $[V_\pi]$ function and Q-learning algorithms.

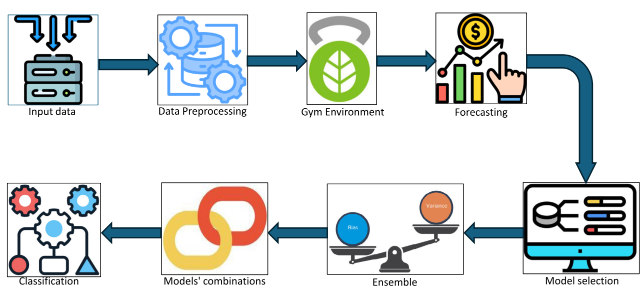

FIGURE 1. Workflow of the proposed ensemble framework.

A. DATA DESCRIPTION

We used the benchmark stock multivariate time series databases such as the Compustat database from Wharton Research Data Services (WRDS) [16], [17], [18], [19], [20], [21], [22], the Chinese stock market database collected from CCER, and the Dow Jones dataset. The overall description of the used datasets is provided in Section VI(A: database description). The detailed description of the subsequent steps is given in the next subsections.

Algorithm 1 Model Selection

1) **Initialization:** Model [M]=[M_1,M_2, ,M_3], sample turbulence (INT), Threshold=T
2) Creating an ensemble method to combine prediction accuracy from each model[M_1 = PPO, M_2 = A2C and M_3 = DDPG] model.
3) Select models based on computed rewards and ensemble these selected models M_1, M_2 and M_3.
4) Dropping the duplicates/ faulty data from data using threshold value (T).
5) Compare each model with other models using the Markov decision process (MDP) technique.
6) INT-threshold(T)=np.quantile(insample-turbulence.turbulence.values,0.90).
7) **Output:** Ensemble result from M_1, M_2 and M_3 models.

Algorithm 2 Working of Different Q-learning Models

1) If (sharpe-PPO $\geq$ sharpe-A2C) and (sharpe-PPO $\geq$ sharpe-DDPG).
2) model-ensemble = model-PPO.
3) model-use.append('PPO').
4) Else if (sharpe-A2C $\geq$ sharpe-PPO) and (Sharpe A2C $\geq$ sharpe-DDPG):
5) model-use.append('A2C')
6) Else:
7) model-ensemble = model-DDPG.
8) model-use.append('DDPG')

B. DATA PRE-PROCESSING

The pre-processing step begins with the utilization of sliding average over moving window w(n). The window length (n) depicts the length of the data components over which the algorithm computes the statistics of multivariate time series data. The sliding average smokes out fluctuations or noise in the multivariate time series data, making underlying patterns more apparent. Moreover, by reducing noise, the sliding average helps to understand and interpret the underlying trends or patterns in the multivariate time series data, making it easier to analyze or build a prediction model.

While smoothing the multivariate data leverages the potential lead for information loss in multivariate time series data, which is highly dependent on time factors, the sliding average over moving window methods employed to preserve the statistical significance information of multivariate time series data components in each sliding data sample and balance between noise reduction and preservation of essential features.

The experimental section provides a detailed analysis of the performance gain attained through this preprocessing step, elucidating the impact on model accuracy and predictive capabilities. The pre-processing techniques are used to remove noises and other artifacts from the MTS data, which is given in detail as follows:

1) **Noise filtering:** The moving average (MA) low-pass filter method mitigates noises from the MTS data. The MA method computes the MTS data (X[N]). It uses former (X[N-i], for [i = 1,$\cdots$, N]) faulty data samples that are associated with dynamic changes and non-stationary characteristics of MTS data from the stock markets [17], [18]. The processed MTS data Y[N] is shown in Eq. 11:

$$Y[N] = \frac{1}{N}\sum_{i=1}^{N} X[N-i] \tag{11}$$

where, N = 5 shows 5-point moving average filtering technique.

C. MODEL SELECTION

The proposed system built a combined model by integrating model [M] = [M_1,M_2,M_3] using ensemble learning techniques. The primary objectives are to improve the overall performance of the prediction accuracy of MTS data analysis and to optimize the computed rewards of the models using the optimized policy of each agent. The computed rewards of each agent are employed based on taken actions in optimal policy using double Q-learning techniques. In the double-deep Q-learning-based proposed framework, the primary objective of the first layer is composed of the policy network. It is to select actions based on the current state $[s_1, \cdots, s_n]$ of the environment for mitigation of overestimated errors and to optimize the influence of error between different layers. The policy network takes the current state as input and outputs the Q-values for all possible actions, from which the action with the highest Q-value is selected. The goal of the second layer in double Q-learning is to reduce the overestimation. It may provide resultant error estimation from the single-layer estimator approach. From Q-values, we predict the next optimized Q value from the policy of the trained agent models (A2C, PPO, and DDPG, respectively. After training the models, we integrate the model of Q-learning on the highest measured Sharpe ratio (SR) and computed reward values for high-level prediction. The proposed system employs an ensemble learning method for decision-making by the selected trained Q-learning agent models (model selection is employed using Markov decision process (shown in Algorithm (1)). To integrate policy-based Q(X, a) and value-based ($V_\pi(s)$), reinforcement learning algorithms are employed the Actor-Critic (A2C) [17] algorithm. For policy-based agents, Proximal Policy Optimization (PPO) [18] was used, which directly learns a policy (a probability distribution of actions) that maps input states to output actions. On the other hand, value-based algorithms learn to select actions based on the predicted value of the input state (s) or action (a). We utilized the Deep Deterministic Policy Gradient (DDPG) algorithm [8] to concurrently learn the Q-function, Q-values, and a policy (π). The working of the different Q-learning models (A2C, PPO, and DDPG) are shown in Algorithm 2. The primary objective for deploying the A2C, PPO, and DDPG algorithms is to use on/off-policy data and the Bellman equation to learn the Q-function and use the Q-function to learn the policy. A2C, the off-policy method, computes the value proposition for each action state (a, s_t) pair for learning agents. On the other hand, the PPO model-free reinforcement learning algorithm provides a powerful policy searchability and allows a learning agent to find the optimal policy by trial and error method. However, PPO leads to high computation and low data efficiency. Moreover, we used the DDPG algorithm to train the model better. It records all the actions and outcomes in a replay buffer and uses the sample to update the RL model by minimizing the loss. This allows the proposed framework for better exploration and faster convergence. Based on the deterministic policy, the double Q-learning model is applied based on improved Q-value as shown in Eq. 12-13). Therefore, once we computed increased Q-values, the policy was enhanced to calculate the maximum rewards (R).

$$R = arg\max_{a} E\left[\sum_{t=0}^{\infty} \gamma^{i} \times R_{\pi(st)}(s_t, s_{t+1})\right] \tag{12}$$

$$Q_{\pi}(s_t, a_t) = \epsilon_{s_{t+1}}[r(s_t, a_t, s_{t+1}) + XX \tag{13}$$

$$XX = \gamma \epsilon_{a_{t+1} \sim \pi(s_{t+1})}[Q_{\pi}(s_{t+1}, a_{t+1}) \tag{14}$$

where the convergence $[0 \leq \gamma \leq 1]$ and policy evaluation are done to compute the state value function. We have used these methods, out of which the model had the best Sharpe ratio, which is considered the classification of the model selected. We have used the Recurrent Neural Network (RNN) and LSTM methods for classification tasks on the same dataset.

D. ENSEMBLE STRATEGY

The ensemble strategy is applied to select the best learning agents based on computed rewards. We used Q-leaning techniques, including PPO, A2C and DDPG learning models, to compute rewards based on selected actions using double Q-learning techniques [19]. The proposed system uses online and off-policy Q-learning techniques for forecasting stock MTS data. Moreover, the Sharpe ratio (SR) method determines the risk-adjusted relative returns from the MTS datasets based on computed rewards. After training these agents, the proposed framework selects the suitable and optimal performing agent model with the highest Sharpe ratio SR) (shown in Eq. 15).

$$SR = \frac{KP - KR}{\sigma_P} \tag{15}$$

In equation (15), KP is the expected return, measured as the risk-free or error rate. $[\sigma_P]$ is the standard deviation of the stock trading data. Based on the computed value of SR, the proposed framework reduced the number of errors and adjusted risk aversion by using an added turbulence index value.

IV. EVALUATION OF THRESHOLD

The primary objective of the evaluation step is to measure the error threshold $[\xi_i]$ that belongs to each faulty and non-faulty data component of a multivariate time series based on a training sample of a given dataset [D]. We estimated the error threshold by modelling the multivariate time series using the ARIMA model. The maximum permissible error threshold $[\xi_i]$ in component $[C_i]$ is shown in equation 16.

A. ARIMA MODEL

To perform accurate and early forecasting of multivariate time series denotes accurate prediction of a class label of data before its complete execution. Moreover, the generation of faulty values or data components may also be available in dynamic multivariate time series due to the deployment of faulty sensors or generating devices or the unexpected occurrence of any artifactss. The availability of faulty data components in MTS deteriorates the performance of the proposed system for the forecasting process of MTS and may provide lower accuracy. Nothing exists in the literature for early classification techniques that may mitigate the faulty data components or provide better mechanisms to handle these irrelevant data components. Hence, these components are mitigated by the MTS data. We solve this issue by training the built model based on significance subsets of MTS. The classification models measure the class discriminating minimum required length (MRLs) utilized for early forecasting of new multivariate faults based on a subset of the multivariate datasets. The proposed system enables the built model to learn an error threshold (using the ARIMA statistical model) for each of the data components in the MTS dataset. We used the error thresholds-based method to identify the availability of faulty data components in new multivariate time series with faulty data components. We measured the errors based on the threshold method during testing. It is used for each data of the MTS datasets using ARIMA models and Q-learning techniques. Overestimated error is measured by Q-learning to train the model to accurately predict faulty components. The error threshold is measured by the ARIMA technique with maximum permissible error threshold $[\xi \in] C_i$ given (Eq. 16) (shown in Fig. 3). ARIMA model employs Maximum Likelihood Estimator (MLE) $(\mathcal{L}(\pm\theta| \pm x) = f(x_T, x_{T-1}, \ldots, x_1; \pm\theta)$ over data [x] $= [x_1 \cdots, x_T]$ (x_T shows the observed [T] MTS data points) to measure a set of ARIMA model parameters $(\pm\theta = \phi_1, \phi_2, \cdots, \phi_p, \phi_1, \phi_2, \cdots, \phi_q)$ is to be estimated from the past time series data (i.e., training data). By modelling the time series, the ARIMA model determines the error threshold (ξ). The highest admissible error threshold ξ in component C_i is given as:

$$\xi_i = \max_{1 \leq j \leq N, 1 \leq k \leq M} \{\epsilon_{jk}\} \tag{16}$$

where $[\xi_{jk}]$ shows an error due to faulty data in the MTS datasets. This error is evaluated by the measure deviation value between k^{th} actual measurement and predicted measurements of j^{th} MTS using the ARIMA model.

B. IDENTIFYING FAULTY COMPONENT OF THE MTS DATASET

The presence of erroneous data in MTS datasets due to different noises and other artefacts. The stock data consists of various technical indicator parameters, highly dependent on the market situation. The noises and dependent variability affect the overall performance of the proposed system. To solve this problem, the proposed system identifies the faulty data component in a new MTS data for accurate classification. Therefore, based on the new MTS training dataset [D] [19], the system can perform as early as possible to achieve maximum benefits. The proposed system classifies faulty and non-faulty data components. The non-faulty datasets correspond to real stock data points) required evaluation of accuracy (α).

Fig. 2 depicts a Hasse diagram for a power set (1,2,3,4). The diagram is divided into two parts: (a) non-faulty data component and (b) faulty component. It displays the Hasse diagram's coloring based on the nodes' relevance. The colors differentiate between highly relevant nodes (blue code node), moderately relevant, and weakly relevant (red color node). The node of the generated POSET graphs consists of two colors representing faulty and non-faulty components, such as the "blue color component" and "red color component," respectively. The white color node shows the most relevant MTS components $[S] = [R_1 \cdots, R_N]$, and "red color" depicts faulty or irrelevant data components of the MTS datasets for early classification of MTS. The proposed framework used the Gaussian Process-based classifiers (GPC) to classify new MTS data components as faulty or relevant (non-fault data components).

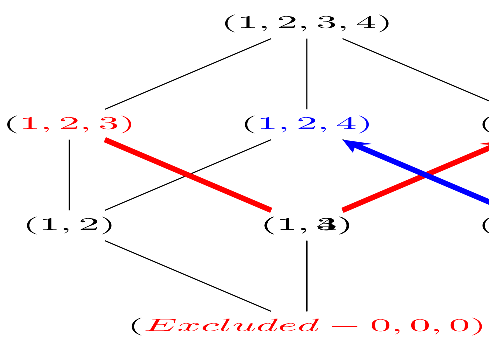

FIGURE 2. Shows Hasse diagram for a power set of $O = \{1, 2, 3, 4\}$.

To classify the MTS dataset, the proposed system generates a Hasse tree of faulty and non-faulty data components and is represented in a partially ordered set (POSET) of lattice graphs (shown in Fig. 2). It helps to sort all the divided datasets into different data components at different labels. The built classification model predicts the classes based on discriminating Minimum Required Length (MRL) from the root of the Hasse tree to the required data points.

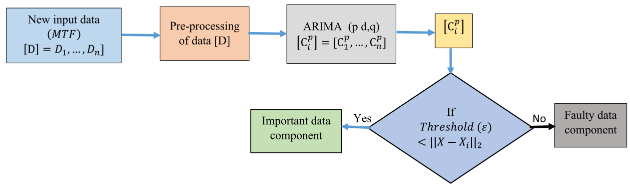

FIGURE 3. Classification of faulty and significant data components in MTS datasets.

C. PREDICTION OF MTS CLASS LABEL

The proposed system employs a Gaussian Process classifier to evaluate the minimum required length of multivariate time series, which is utilized to predict a class label of new multivariate time series with irrelevant data/ faulty data in MTS. The measured minimum needed length of MTS is considered to evaluate the dependency of data components in a small subset of datasets using the recurrent neural network (RNN) model. Gaussian process-based classifier excels at capturing complex relationships between variables, while a proposed system with RNNs is proficient at modelling temporal dependencies over sequential data points. Therefore, the proposed system employs classification models $[K_s]$ enabled GPC-based function (hm_i) to predict the class label $(L_i^{''})$ as faulty data or non-faulty data using $C_i^{p,s}$, where m_i refers the number of MTS data points in $C_i^{p,s}$ at the time of accurate prediction. Here, $[m']$ is interpreted as the expected learned agent values needed to predict a class label $[L]$. The identification of faulty data is given in Algorithm 3 and Algorithm 4. The learning phase of the proposed model is illustrated in Algorithm. Based on overall observations, the proposed system enabled with GPC and RNN provides valuable assistance to the proposed framework using Q-learning by enriching the model's predictive capabilities.

The primary motivation of this study is to create a novel framework enabled with Gaussian process classifiers (GPC) and Q-learning's recurrent neural network (RNN) for forecasting dynamic multivariate time series prediction lies in their complementary capabilities. Gaussian process classifiers are leveraged with probabilistic models to capture complex relationships within multivariate data by modelling the distribution over functions. The proposed framework, enabled by GPC, performed well and effectively handled uncertainty and non-linear relationships. It is well-suited for capturing the intricate patterns often present in multivariate time series data in different time internals. The proposed framework integrates GPC and RNN models for multivariate time series prediction to improve the ability to capture global multivariate trends and local stock fluctuations within the time series data. The proposed framework with the GPC classification model provides complex data patterns and intricate relationship representations between variables.

Algorithm 3 Prediction of Class label

1) Perform computation $m' =$ min (M1) and L= $arg\,\max_q$ min(M1).
2) **For** iteration k=1,2,... s **do**
3) If $m_0 \leq m_k$
4) then perform prediction of the class labeled of $C^{p,s}$,
5) Else collect data points,L^i shows a predicted label.
6) If labelled are matched with new data points (L^i= L)
7) Update m_0=(M_{i+1} |L^i) for the next component and L=l.
8) Else
9) Measure the MTS length to mitigate the faulty data.
10) Reduce the effect of the inaccurate prediction of MTS and go to step 2.
11) Search a class label (l) predicted by the MTS components and set it to C^{ps}.
12) Early forecasting is accomplished in the next phase using the number of data points for the prediction.

Algorithm 4 Model Learning Phase

1) **Initialize:** Labeled database (D) includes N data component with labels (l_n) [n=0 faulty, and n=1 (non-faulty)] components of complete dataset (MM).
2) **Output:** A set of classifiers [K] = [K_1, K_2, K_3]
3) For k=1 to K models where (k=1,2 3)
4) Compute scores on MTS database(MM)
5) Sort the trading point components in increasing orders
6) Arrange data in non-linear binary tree data structures.
7) Obtain the relevance data component (R) =[$R_1, \cdots, R_P$].
8) For i= 1 to P (from all MTS components and (j=1 to S)
9) Apply the k-means clustering method to get the confusion matrix for clustered data components.
10) f ← i=0 to MM (MTS data component)
11) **Do** Build a Gaussian process classifier [K]=[$K_1,\cdots,K_r$] where K_i=(W_i,R_i] for 1≤ I ≤ r for faulty data.
12) Evaluate error thresholds for [i ←1] to n do
13) Compute error threshold using ARIMA model.
$$xi = max_{1\leq j\leq N} 1 \leq K \leq M(\xi_{jk}) \tag{17}$$
14) where the error thresholds [ξ] =[$\xi_1, \cdots \xi_k$]
15) Return K, and ξ

At the same time, customized RNN techniques give a better understanding of temporal dependence due to the occurrence of dependent and independent data points and variables in the multivariate time series data over sequential data points. The proposed framework combined GPC and RNN models to provide valuable assistance to Q-learning by enriching the model's predictive capabilities. GPC contributes by providing probabilistic predictions, enabling Q-learning to make informed decisions while considering uncertainty. RNN, on the other hand, aids in capturing temporal dynamics and dependencies, enhancing the predictive accuracy of Q-learning over time-varying data. The proposed framework computes the relevancy of the MTS using k-means clustering techniques and arranges these measured scores in a non-increasing order based on the significant scores. The computed scores measure overestimated errors at the second layer of the proposed system-enabled Q-learning based on the required discriminatory lengths and classification of unknown multivariate time series leveraged faulty data components using Q-learning techniques.

D. RETRIEVING RELEVANT SUB-DATASETS USING POSET

The proposed system performs the computation to achieve the essential sub-datasets from the given MTS data (D) using the partially ordered sets (POSET) Hasse graph theory method. The suggested system groups faulty and non-faulty MTS data using the k-means clustering approach to calculate the critical components from collecting all MTS components. It determines these components' relevance scores based on a rank calculation. The rank of these scores of each data component provides a strategy to select discriminatory feature components from clustered data, and the relevancy score is calculated as (shown in Eq. 18).

$$S\,(P, Q) = \sqrt{\sum_{i=1}^{n} (Q_i - P_i)^2} \tag{18}$$

$S\,(P, Q)$ measures the L2-norm distances between POSET Hasse graph node values that are clustered and obtained by applying the k-means clustering technique on faulty (P) and non-faulty (Q) datasets, respectively.

Now, the optimal data components of the MTS are arranged in non-increasing order in the max-heap data structure to find the better representation for achieving node relevancy scores to construct Hasse graphs (shown in Eq. 18).

Based on measured relevancy scores, the proposed system builds a Hasse graph, which is POSET of all possible subsets (i.e., 2^n, where (n) is the total number of the components). In the Hasse diagram, the proposed system estimates each node's significance based on relevant scores. It accordingly provides a color-coding scheme for the faulty node (P) (red color) and non-faulty (Q) (white color). The colour-coding process signifies that if the node is white, it is relevant data components; otherwise, the node with a 'red' colour signifies faulty data components. The selection of a node depends on measured values. It is said to be relevant with C_i^p (where p = (P, Q) components based on computed scores by using probabilistic ARIMA model if (accuracy $(A)_{C^P} \geq \alpha$ accuracy $(A)_{C^Q}$ holds, where (α) denotes the required level of accuracy (A) for finding faulty and non-faulty data component using the ARIMA model.

E. MRL COMPUTATION

We employ a GPC and k-means clustering methods to estimate the required length of data components for discriminating MTS data components (S) = [$R_1, \cdots, R_N$] with a subset (s) ∈ (S).The Gaussian process is a non-parametric model on Gaussian probability distribution functions. It adapts to different MTS patterns and complexities in the data without explicitly pre-defining the functional form. The GPC implements the Gaussian processes model for classification. To classify the unknown MTS data, we built a GPC model to find the faulty components in the time series data [20], [21]. Here, we defined [$X_f \in S_i$] as a multivariate stock time series with stock data points [D]. In this step, the proposed framework used the GPC (hm_i) to model the MTS classification as a stochastic process outcome and receive class posterior probabilities using the given dataset S_i. Let (hm_i) is a Gaussian process classifier (hm_f): [$R^D \rightarrow$ L]. We measured the posterior probability of stock class [L_q] for the stock trading data. The evaluation of the posterior probability [X_D] belongs to k-means clusters (G_q) which is measured by using Baye's rule:

$$p(G_q) = \beta[q][q] \tag{19}$$

$$prob.(G_q|X_D) = \frac{D_q}{\sum_{K=1}^{L} D_k} \tag{20}$$

where $D_q = G_q$-D_k represents the average of Euclidean distances from X_D to the total number of obtained k-means clusters (L). D_q depicts the average distance from X_D to cluster G_q, and is given as:

$$D = \frac{1}{L}\sum_{q=1}^{L}(X_D, G_q)\&D_q = distance(X_D, G_q) \tag{21}$$

The function distance (X_D,G_q) utilizes only initial [D] trading data points of stock of the clusters to evaluate the similarity scores.

F. COMPUTATION OF MINIMUM REQUIRED LENGTH (MRL)

In this section, we defined [X_D] $\in S_i$ a MTS with data points [D$\leq$ M], where D is less than MTS data points [M]. The necessity of calculating the Computation of Minimum Required Length (MRL) for multivariate time series data for prediction lies in its role within the proposed framework, which comprises two major phases: learning and testing. In the learning phase, the framework constructs a series of classification models using a provided training multivariate time series dataset. Each model is designed to learn class-discriminating MRLs (Minimum Required Lengths) and error thresholds corresponding to the components of the time series data. This learning process allows the models to understand the optimal length of data required for accurate classification, as well as the thresholds for distinguishing between different classes or states. Once the models are trained, they leverage the learned MRLs during testing to classify a new dataset with faulty components, denoted as MTF (Multivariate Time Series with Faulty components), corresponding to an ongoing multivariate time series prediction. By utilizing the learned MRLs, the models can assess and classify the faulty components in real-time or before the complete execution of the prediction process. Calculating the Computation of Minimum Required Length (MRL) is essential for the framework as it enables the classification models to accurately identify and classify faulty components in multivariate time series data during the testing phase.

There is a need to identify the faulty data component in the MTS data for accurate and early classification. In this step, the proposed framework used built GPC ((hm_i) to the model of the MTS to measure the faulty component as the outcomes from the data and calculated posterior probabilities based on the given dataset [D_i]. The GPC estimates the class label (L) using (C_i^p), where m_i shows the number of data points in (C_i^p). Here, ((hm_i) depicts a GPC where mapping and interference (hm_D): $R^D \rightarrow$ L). The GPC computes the maximum posterior probability of a given class L_q (shown in Eq. (22-Eq. 26):

$$hm_f = Prob.(L_q|X_f) = \prod_{k=1}^{f} \frac{p(L_q).p(x_f^k|L_q)}{p(x_f^k)} \tag{22}$$

where $p(x_f^k)$ and $p(L_q)$ are marginal, and prior probabilities are used to compute the minimum required length from MTS for faulty and non-faulty estimation using S_i.

$$p(X_f^k|L_q) = \frac{1}{\sqrt{2\pi\sigma^2}}e^{\frac{(x_f^k-\mu^2)}{(2\sigma^2)}} \tag{23}$$

The statistical measures are defined for [D_f] to determine the relative accuracy, which is achieved using (f) stock data points instead of using all MTS data points [M]. The primary aim of the GPC classifier is to learn the data mapped [X$\rightarrow$L] using a given dataset (D). The term [L] depicts the ground truth class label, and [f $\leq$ M] is the required minimum stock data (shown in Algorithm 5).

The evaluation of the maximum likelihood (defined prior probability) of the given (L_q) class label is given in (Eq. 19). The estimation of the probability of a new time series [D] with a total number of faulty data points shown in (Eqs. 20-26): Using Eqs. (19), (20), and (21), the average accuracy of stock data [X_f] is defined. The ground-truth values of the given class label are given as follows:

$$Y_f = \frac{prob.(L_q X_f) \times p(G_q, D_f)}{\alpha \times p(G_q)} \tag{24}$$

where (α) shows the required level of accuracy obtained by the previous data history. The trade-off between an average measure (F_f). The term (ϵ) depicts the earliness to classify faulty data components based on defined utility function [$U(.)$]. This optimizes the faulty error for obtaining the maximum level of forecasting accuracy given as follows (shown in Eq. 25)

$$U_{Xf} = \frac{2 \times F_f \times \epsilon}{F_f + \epsilon}, \epsilon = \frac{[M-f]}{[M]} \tag{25}$$

After that, the minimum required length (MLR) of the stock time series X *in* S_i is measured as follows (shown in Eq. 26)

$$MRL = argmax_f[(U(X_f])] \tag{26}$$

Algorithm 5 Testing Phase

0: **Initialize:** We defined (K) classifiers, error thresholds, and MTS data.
0: **Output:** Predicted class label (L).
0: fmin(M_i=m_1) then
0: **While** (C_p) is not completed
0: **Do Check** for all models for i ←1 to n **Do**
0: **Build** ARIMA model for (C_i^p) at time (t)
0: **Perform** prediction of next data point (x^{t+1})
0: **If** the value ($\xi \leq ||x - x^{''}||_2$) then
0: **Detection** System detect the *Cpi* faulty data component and remove inaccurate data from C^p
0: **Choose** next classifiers (Ks ∈ K) for α.
0: **Choose** the data components from $C^{p,s}$ needed for classification models K_s
0: **Evaluate** $m^{'}$ =min (M1) and L= $argmax_q min(M1)$
0: **For** i ← 1 to s **Do**
0: **If** $m_0 \leq m_i$
0: **Prediction** After, we perform the prediction of the class label of $C_{s_i}^p$.
0: **Else** Collect time series related data. Let L^i show a predicted class label.
0: **Predict** label L of $C_{s_i}^p$ using GP classifier.
0: **Else** store mote MTS data and go to the next step (2).
0: **If** l= = L then
0: **Measure** m_0=(M_{i+1} |L^i)
0: **Append** new class label into previous data.
0: **Then** Go to line step
0: **Evaluate** $m^{'}$.
0: L= arg max$_q$ min(M1)
0: **End** If
0: **End** for i
0: iteration=iteration+1
0: **End** While
0: Return L

V. RESULT AND DISCUSSION

The experimental results of the proposed approach are evaluated on benchmark time-series datasets.

A. DATASET DESCRIPTION

The experimental results of the proposed approach are evaluated on benchmark time-series datasets. The stock database is taken from open sources to assess the proposed model. The performance evaluation is based on the stock Compustat database through the Wharton Research Data Services (WRDS) [22]. Q-learning agents such as PPO, A2C, and DDPG are used to learn new data in the trading and training phases. Next, a validation stage is used to validate the used model and learning agents based on the Sharpe ratio [22].

Finally,the proposed model evaluates the profitability of each ensemble technique based on deep Q-learning algorithms. We have divided the whole dataset for further processing and analysis,as shown in Fig. 4. The LSTM model is used to find the actual and LSTM values from the split stock for training the models given in Fig. 4. Fig. 4 depicts the training dataset and test datasets from dataset-I (see Table 1). On the other hand,dataset-II comprises nine stock trading data from the Shanghai Stock Exchange dataset (shown in Table 2).

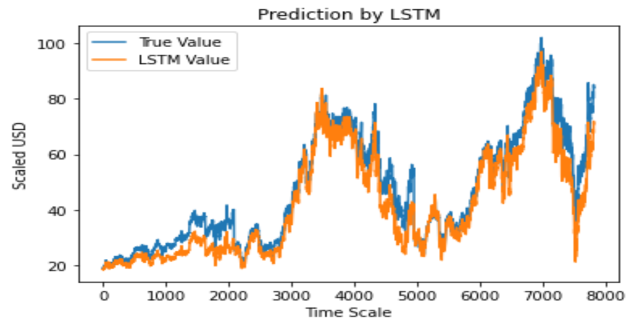

FIGURE 4. Prediction of stock data using LSTM method.

TABLE 1. Training data and test dataset-I.

SD	Training period	Testing period	BCT (Thc, tmsr)
Apple	2008/01/02-2008/12/30.	2009/01/02-2009/06/30.	(0.0010,0.0060)
AUO	2006/01/02-2008/10/14.	2008/10/15-2009/04/09.	(0.0010,0.0060)
UMC	2006/01/02-2008/10/14.	2008/10/15-2009/04/09.	(0.0010,0.0059)
BA	2008/01/02-2008/12/30.	2009/01/02-2009/06/30.	(0.0016,0.0064)

In Table 3,we used the training period,testing period,and Bi-cluster threshold parameters to estimate early prediction. The values of parameters are determined by applying the grid search method,where stock data (SD),bi-cluster thresholds BCT) (Thc,Tmsr) = (value1,value2). The range of the BCT values lies between (Thc,Tmsr)∈[0-1]). The step size varies from 1 to 25 in steps of 5. The proposed system requires training,and at least the batch size (50 features) and 50 epochs ar optimized with the number of required parameters for model neurons.

We used the grid search method to find the discriminatory values with step size what is the search step size,and how they are accurately determined? Please explain in the article.

We also used the Chinese stock market datasets, IBM, Walmart, Citigroup, DowJones, and S& P stock datasets for the performance evaluation of the proposed framework. The primary distribution of sample data patterns of the Chinese stock market is shown in Table 3.

B. EVALUATION ALGORITHMS

We utilized a diverse set of ensemble learning techniques to assess the performance of our system,including the support vector machine classification approach. In addition,we leveraged the power of ensemble random forest and gradient

TABLE 2. Training data and test dataset-II.

SD		TP	TEP	textbfBCT (Thc, tmsr)
Down trend	600197	2010/01/04-2010/11/18.	2010/11/19-2011/08/18.	(0.0019,0.0094)
	600211			(0.0031, 0.0104)
	600488			(0.0024,0.0120)
Steady trend	600066			(0.0027, 0.010)
	600697			(0.0029, 0.0157)
	600881			(0.0010, 0.0060)
Up trend	600051			(0.0014,0.0074
	600107			(0.0028,0.0081)
	600167			(0.0137, 0.0099)

We used these parameters, which are determined by grid search, TP=Training period, TEP=Testing period, SD=Stock data, BCT=Bi-cluster thresholds

TABLE 3. Distribution of 13 patterns of stock trade forecasting analysis.

Pattern	No	Ratio	Ratio 1	Ratio 2
0	0.7776	0.10%	0.10%	0.60%
1	15.269	0.21%	0.23%	0.13%
2	56.477	0.77%	0.78%	0.79%
3	57.910	0.79%	0.82%	0.69%
4	12.549	0.16%	0.17%	0.20%
5	7.501	0.10%	0.12%	0.05%
6	107.234	2.33%	2.35%	2.31%
7	630.482	8.63%	8.65%	8.71%
8	247.082	3.38%	3.54%	2.75%
9	186.263	2.66%	2.68%	2.78%
10	563.830	7.72%	7.60%	8.89%
11	2442.454	33.45%	33.50%	34.88%
12	2905.003	39.95%	39.88%	39.99%

decision-boosting tree-based classifiers to handle both continuous and discrete time series data with ease. We employed cutting-edge LSTM logistic regression (LR) and ARIMA models to further enhance our system's predictive capabilities. Furthermore,we pushed the boundaries of ML by implementing ensemble Q-learning (QL) algorithms to train various agents and gauge their performance against a stock database. We evaluated these agents based on their response to the Sharpe ratio (SR),ultimately selecting the most effective QL-based agent for active trading. In summary,our approach harnesses the latest techniques in ensemble learning to deliver accurate and reliable predictions for trading in the stock market. By combining a range of algorithms,we have created a powerful and sophisticated system that can handle diverse data types and produce superior results.

1) **Performance evaluation on Sortino Ratio (SR):** Sortino Ratio (SR) is a statistical parameter metric used to assess a trading mechanism and evaluate performance. It provides a better observation of expected returns from the stock market and online trading systems based on the minimum involved market risks. The computation of SR is shown in Eq. 27).

$$SR = \frac{R - r_f}{\sigma_d} \tag{27}$$

The term R shows the return based on selected aggregated daily,weekly,monthly,or yearly. The term r_f estimates the associated target value or rate and (σ_d) shows the returns from the downside deviation of markets,which measures negative returns based on the calculatedstandard deviation and only determines the estimated strategy's volatility (or risk). The SR insignificantly varies from the SR metric. It needs a standard deviation on all return values and does not consider the non-positive values.

We used both metrics to evaluate the accuracy based on selected statistical models (ARIMA and other classification models). The given model by Moody and Saffell [12] maximizes risk-adjusted returns rather than profits by considering such a metric. Therefore,we defined the objective to measure statistical parameters based on the mean and standard deviation as

$$Max_\theta = \frac{mean[(R_1, R_2, \cdots R_N)]}{std[(R_1, R_2, \cdots R_N)]}|(\theta) \tag{28}$$

The term (θ) shows the family of statistical parameters of used models. We customized our proposed model based on a selected neural network with defined parameters $(\theta) = (w, b)$. The parameters $[b]$ and $[w]$ are selected to train our model to maximize the defined objective for better rewards for learning Q-learning algorithms (shown in Eq. 29).

$$max_\theta \sum_{t=0}^{T} R_t|(\theta) \tag{29}$$

TABLE 4. Average accuracy of statistical ML method.

Feature set	Model	MASE
150	ARIMA	0.7776
200	SVM	0.8278
350	K-NN	0.8116
400	DT	0.8926
500	ANN	0.8654
600	MLP	0.8754
700	LSTM+CNN	0.9054
800	RBF	0.9654
900	**Prop.**	0.9875

C. PERFORMANCE EVALUATION ON STATISTICAL MEASURES

We used the non-parametric Friedman rank-sum testing method to evaluate the statistical differences in the compared forecasting methods. Also,we applied Student's t-test and Hochberg's post-hoc procedure to examine these differences further. The symmetric mean absolute percentage error (SMAPE) and mean absolute scaled error (MASE) measures are employed to accomplish the statistical testing,with a significance level of $(\alpha) = 0.05$.SMPE method measures to evaluate the accuracy of forecasts of multivariate time series. The primary motivation for the proposed MASE mathematical formula for estimating new faulty components in multivariate time series data is that it offers a normalized measure of prediction accuracy relative to a baseline forecast. By considering both scale and direction,the proposed MASE formula provides a robust indicator for evaluating the performance of forecasting models based on with seasonal patterns of faulty and non-faulty data components. Its application in this context facilitates effective assessment of the accuracy of predictions,aiding in detecting and classifying faulty components within the time series data.he equations (30 and 31) can be rewritten as follows:

$$\mathrm{SMAPE} = \frac{2}{N}\sum_{t=1}^{N}\left(\frac{|F_t - Y_t|}{|F_t| + |Y_t|}\right) \tag{30}$$

$$\mathrm{MASE} = \frac{1}{N}\frac{\sum_{t=1}^{N}|F_t - Y_t|}{\frac{1}{n-S}\sum_{t=S+1}^{n}|Y_t - Y_{t-S}|} \tag{31}$$

where, $[Y_t]$ shows the observation over time [t], and the rendered forecast is depicted by F_t. The terms [N] and (n) denote the number of MTS data points in the test dataset and the total number of observations in the training data set of an MTS dataset.

$$Earliness(\%) = \frac{M - MRL}{M} \times 100 \tag{32}$$

F1 Score: The F1-score is defined to study the impact of early classification and earliness detection on the accuracy of the early classifier. The F1-score is defined mathematically and expressed as

$$F1 - score = \frac{2 \times accuracy \times (1 - \frac{MRL}{M})}{(1 - \frac{MRL}{M})accuracy} \tag{33}$$

$$\text{G-mean} = \sqrt{\text{Sensitivity} \times \text{Specificity}} \tag{34}$$

D. PERFORMANCE ANALYSIS

The performance analysis of the different classification models such as ARIMA, K-Nearest Neighbour (K-NN), Decision Tree (DT), Artificial Neural Network (ANN), and Long short-term memory (LSTM) and Radial Basis Function (RBF) models are evaluated based on the MTS database using the 10-cross validation settings (shown in Table 4 and Table 5, respectively).

The performance of statistical ML models is measured based on selected discriminatory features (shown in Table 4) based on the computed minimum required length of feature vectors observed over the time interval [t] over forecaster data (shown in equation 32). The feature points change the measured time intervals in MTS data analysis. The results show inconsistent characteristics due to the non-availability of a discriminatory set of features. Moreover, the classification models fail to measure the class discriminating MRLs as a feature set for early and accurate prediction of the new multivariate time series with irrelevant or faulty data components(shown in Table 4).

We considered the dynamic changes in the MTS data values in a Markov Decision Process (MDP) based environment that represents the system or process generating the time series data. It also includes historical data, MTS data trends, complex data patterns, and any external factors that may impact the time series. In Table 5, we considered the randomness generation in the commutative rewards (R) and discount factor $\gamma \in [0.1 - 0.95]$ in all experiments where the learning rate as linear $\alpha_t(s, a) = \frac{1}{n_{[t]}}(s, a)$ updated Q-values for each iteration. The proposed framework-enabled double deep Q-learning model incorporates a generated network for producing the Q-value for every Q-learning algorithm, and a target network is used to produce the target Q-values to train these parameters. We used the rectified linear units (ReLU) non-linear activation function in the double-deep Q-learning model to mitigate the gradient vanishing problems.

TABLE 5. Shows average accuracy of different models for classification.

Model	Accuracy (SMAPE) (%)	AUC(%)
LR	0.5566	0.6167
MLP	0.7845	0.8754
ARIMA	0.6896	0.7984
CNN+SVM	0.8594	0.9098
Q-Learning+SARSA	0.8594	0.9098
Q-learning+TD	0.9198	0.9389
Prop.+ARIMA+DQL	0.9689	0.9985

Abbreviation:LR= Logistic regression, MLP: Multi-percepton layer, CNN= Convolutional Neural Network, and TD= Q-learning based on Temporal Difference learning method, DDL= Double Q-learning.

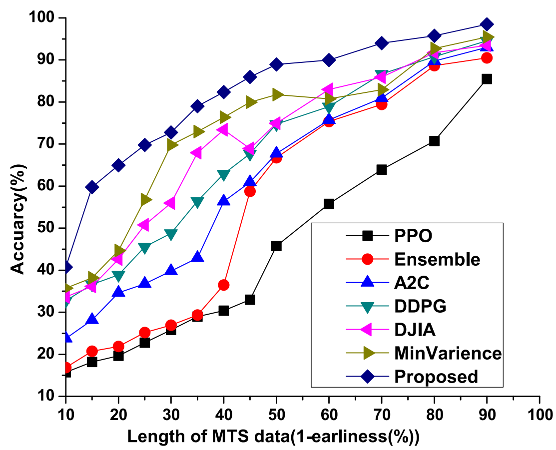

FIGURE 5. Accuracy of different techniques with the proposed method.

Fig. 5 shows the cumulative return trade-off between the stock prices using PPO, Ensemble, A2C, DDPG, DJIA, and Min-Variance methods. Fig. 6 measures the commutative returns and accuracy of the proposed system based on buy and hold, long and short, and full buying and selling of items in the stock markets. The system evaluates the accuracy and its impact on the selling and buying of electronic products in the markets. The variances are measured based on different thresholds to analyze the market ups and downs in different time intervals. The performance evaluation of reinforcement learning agent is shown in Table 4). Table 4 shows the

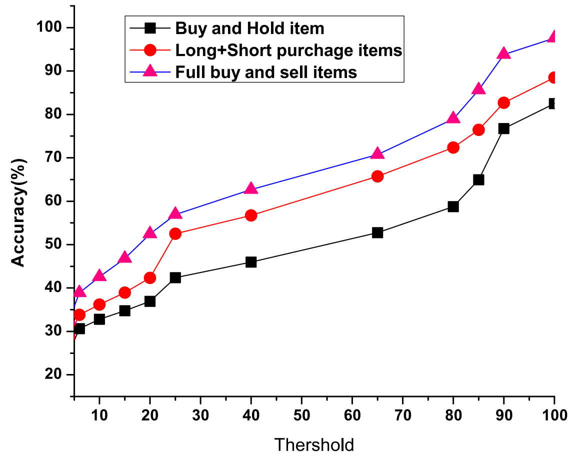

FIGURE 6. Cumulative returns on buy-hold, long+short buy and selling and full buying and selling condition of products in stock markets.

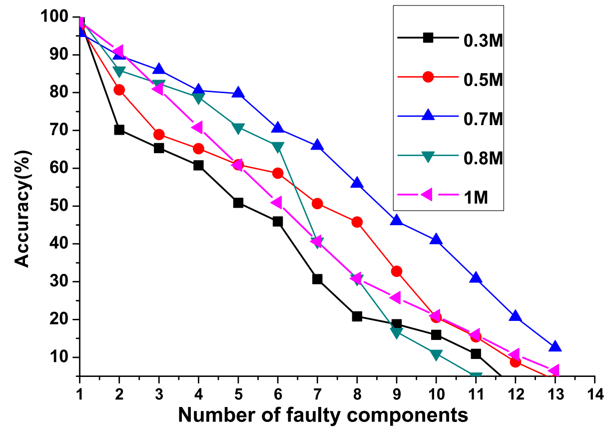

FIGURE 7. Accuracy vs several faulty component length (M) of MTS dataset.

measured Sharpe ratio where the value of the Sharpe ratio is 1, 0.5, and 0.23. The value of the Sharpe ratio greater than 1.0 shows that it is acceptable for investors to invest in stock markets.

TABLE 6. Performance evaluation of learning agents.

Method	Validation Interval	Sharpe Ratio
A2C	20171003-20180103	2
PPO2	20200906-20200406	0.50
DDPG	20200106-20200406	0.23

E. FAULT TOLERANCE CAPABILITY OF PROPOSED FRAMEWORK

We experimented to validate the proposed framework's system performance and fault tolerance with varying numbers of faulty components in the testing MTS data. The experiment uses a training dataset with the total MTS dataset. Based on the achieved relevancy order of data components using the POSET lattice graph, the framework is evaluated for the following cases:

1) **Faulty components taken from high order relevancy:** Fig. 7 shows the capability of the fault tolerance of the proposed framework with variances of selected faulty components from the four datasets (i.e. HAC, HHAR, DSA, and NTU RGB+D. The different lengths of the MTS dataset are given as 0.5M, 0.6M,0.8M, and M, where M shows the selected complete length of MTS datasets, and 0.5M shows 50% length of the MTS for measuring the fault tolerance for evaluation of experimental results.
   Fig. 7 shows the calculated returns based on the selected optimal policy by double Q-learning agents. It depicts that the returns are considered with increasing drawdown for better investment of the money in stocks.
2) **Multiple combinations of DQL decisions:** We combined the decisions from multiple learning agents to find the optimal policy for achieving a maximum reward. To reach decisions from multiple double Q-learning (DQL) agents, we trained these learners based on available datasets using pre-trained models at different epochs in the defined environment. The proposed system used RL classifiers for selecting their votes based on bagging and boosting the ensemble learning strategy for each training epoch [23].

F. PERFORMANCE EVALUATION ON SELECTED FEATURES FROM FAULTY AND NON-FAULTY MTS DATA

The performances of the proposed framework are evaluated to find the relevancy of the data components of MTS after pre-processed datasets. We measured the faulty and non-faulty components of MTS data for detailed analysis based on clustered selected feature sets as vector components using different datasets by taking faulty components and selecting features from the low end of the relevancy order. For example, n = 15, if the a number of faulty components is 7, then it means data components (extracted features) of MTS C15, C14, C13, C12, C11, C10, and C9 are faulty (shown in Table 7 and Table 8, respectively.

TABLE 7. Performance evaluation on pre-processed data components of MTS.

Method	**Feature set**	Accuracy (%)
Logistic Regression	100	76.95
K-means	250	73.90
DDPG	350	79.98
Random forest	550	82.93
Decision tree	650	85.99
SVM	750	88.98
Prop.	850	96.78

The results from the HAR, DSA, HHAR, and NTURGB databases, as well as the proposed technique, are depicted in Fig. 8. The proposed method's earliness (%) is $\alpha = (0.5, 0.6, 0.7, 0.8, 0.9)$. It is accomplished by employing the Eq.(32) set of sample rates for the datasets employed. The results show

TABLE 8. Average accuracy(%) based on extracted features with G-mean and AUC.

Model	Accuracy	Loss	Sensitivity	Specificity	G-mean
DT	90.89	12.93	94.98	0.08	0.2907
SVM	89.99	23.98	0.94	0.86	0.9185
K-NN	82.78	36.91	0.94	0.85	0.8944
IDT	95.84	35.96	0.99	0.95	0.9718
Ensemble	98.99	76.80	0.98	0.94	0.9608

Abbreviation: DT = Decision Tree, IDT = Incremental Decision Tree, SVM = Support Vector Machine, K-NN = K-Nearest Neighbourhood

that the suggested techniques outperform existing approaches with varied (α) settings on multiple datasets. We compute the earliest and most accurate class label for each class label as the MTS progresses (see Fig. 7 and Fig. 8). The suggested system's accuracy is calculated using Eq. (refeq:99). The number of MTS data points (%) that are not used in the prediction is regarded as the earliest.

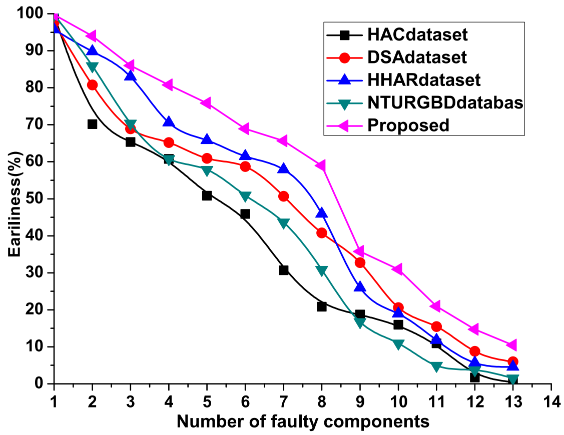

FIGURE 8. Earliness (%) of the proposed system based on a selected number of faulty components of the different existing datasets.

G. COMPARATIVE ANALYSIS WITH EXISTING METHODS

In this section, we compared the performance of the proposed framework with the existing approaches for the classification of the MTS stock trade time-series data. For comparative analysis, we considered forty-one patterns for early classification using double-Q learning and different ML models. Based on overall observations, the comparative result reveals 96 forecasts shown in Fig.8, and Fig. 9.

In Table 10, the proposed framework used an ensemble strategy to yield a Sharpe ratio (SR), which varies from 1:25. The SR ratio determines the powerful values representing time to lend money for a long moment in stock prices. The proposed framework presents higher accuracy than the SR with 0:53, and other methods with SR = 0:65 provide better results based on min-variances. The annual return and the annual volatility measure the system's capacity for accurate prediction and dynamic changes in the stock market due to slight ups and downs. Based on overall observations, it can be concluded that the proposed framework-based ensemble strategy gives a much higher annual return, and the annual volatility is lower than others.

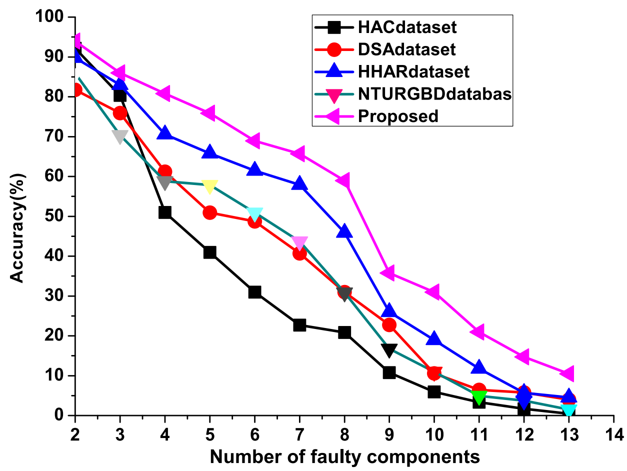

FIGURE 9. Accuracy (%) of the proposed system on several faulty components of the different existing datasets.

TABLE 9. Distribution of the train and test data sample of the existing datasets.

Dataset	N	N_{train}	N_{test}	n	M	l
HAR	24000	16800	7200	11	900	8
DSA [25]	3360	2352	1008	15	125	7
HAAR [24]	20000	14000	6000	12	1200	6
NTU RGB+D [25]	8000	5600	2400	9	1000	10

H. PERFORMANCE EVALUATION ON EXISTING DATSETS

The performance of the proposed framework is evaluated on public datasets for comparison with the existing methods.

1) **Human Activity Classification (HAC) database [24]:** An experiment is done to collect a real-time human activity dataset.
2) **Daily and Sports activity (DSA):** MTS data on 19 different types of human activities with ground truth class labels are included in the DSA [25] database. An MTS is formed by the actions recorded by several intelligent sensors.
3) **Heterogeneity HAC (HHAR)+NTU RGB+D [25]:** The HHAR dataset is built to understand the significance of collected mobile phone sensor data in different time intervals. The proposed system uses 70% of the total datasets for training the model, and 30% datasets are used to test the model based on a 10-fold cross-validation method (shown in Table 9).

TABLE 10. Comparison of existing techniques with the proposed method.

P	Prop.	[8]	[16]	[18]	[15]	[14]	[4]	[22]
CR	80.86	84.89	74.95	65.89	35.98	39.75	10.75	42.50
AR	14.56	16.56	13.78	11.67	6.78	5.68	4.32	3.19
AV	8.76	13.6	11.67	12.89	18.8	17.75	16.62	20.15
SR	1.25	1.10	1.21	.72	.65	.53	.47	.47.79
Md	-8.23	-24.89	-10.89	-14.75	-25.79	-36.78	-25.78	-35.98

Abbreviations: Prop. = Proposed Ensemble Method, P=Parameters, CR= Cumulative Returns (Dollar), AR(%)= Annual return, AV(%)=Annual Volatility, SR=Sharpe ratio, Md(%)=Max-down, M=Min-Variance

TABLE 11. Comparison of the proposed method on stock datasets.

SD	ANFIS-RL	Prop.	BCL	Diff.	R
IBM	212.27	235.88	T0=0.001, $T_n w$=0.006	+23.89	2
WT	110.69	238.19	T0=0.001, $T_n w$=0.005	+127.56	5
CT	131.68	105.38	T0=0.001, $T_n w$=0.006	-27.46	1
DW	193.98	216.48	T0=0.001, $T_n w$=0.006	+67.96	5
S&P	187.72	245.98	T0=0.001, $T_n w$=0.005	+57.76	4
AV	177.89	215.38	T0=0.001, $T_n w$=0.004	49.9426	3

Abbreviations: SD=Stock datasets, CT=Citigroup,WT= Walmart, DW=Dow Jones, AV=Average, Prop.= Proposed Ensemble method, BCL= Bi-cluster thresholds, Diff.=Differences, R=Rank. These parameters are measured based on the grid search method.

VI. CONCLUSION AND FUTURE DIRECTIONS

A novel system is proposed for forecasting stock trade points using the Q-reinforcement-based ensemble methods and probabilistic ARIMA models. We used ML techniques such as support vector machines, random forest, boosting methods, and other classifiers for prediction and inaccurate data classification. We compared the performance of the proposed systems with the existing methods, including k-NN, support vector, decision tree, ANN and MPL, LSTM, and other techniques. We classified MTS datasets even when they had some defective components. The proposed framework constructs a set classification model using a GP classifier by estimating the class discriminating minimum required lengths of data components. The classification model uses the ARIMA model to classify the faulty data from MTS components in the new MTS datasets. This work prompts further research to promote new avenues for early classification of accurate data based on extracted feature basis and involving disruptive methods based on early classification for time series data.

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